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[23MSCH1H02]

FIRST SEMESTER UNIVERSITY EXAMINATION, JANUARY - 2025

ORGANIC CHEMISTRY - I

Time: 03 HOURS **Maximum Marks:50**

		Blooms Level	CO	Marks
	SECTION – A			
1.	**Answer any FOUR of the following questions.**			**4×5=20**
a.	Explain the sequence rule for specification of the Cahn-Ingold-Prelog *R/S* system.	2	1	**05**
b.	What are the factors affecting the stability of carboanions? Explain.	2	3	**05**
c.	Illustrate the energy profile diagram for SN1 and SN2 reactions.	3	4	**05**
d.	Classify Friedel-Crafts alkylation and illustrate the stepwise mechanism involved in the alkylation of benzene.	4	5	**05**
e.	What is aldol condensation? give an example with a mechanism.	3	6	**05**
	SECTION – B			
2.	**Answer any TWO of the following questions.**			**2×9=18**
a.	How do enantiomers exhibit optical activity and expain with the suitable example.	3	2	**09**
b.	**i.** Explain isotopic labeling with suitable examples. **ii.** Give any five methods of generation of nitrene	3	3	**09**
c.	Examine Markownikoff's and anti-Markownikoff's products, providing suitable examples to illustrate their differences.	3	4	**09**
	SECTION – C			
3.	**Answer the following questions.**			**1x12=12**
a.	**i.** Define stereoselective and stereospecific reactions with suitable examples.	3	1	**04**
	ii. Discuss the mechanism of E1 elimination.	2	5	**04**
	iii. Write mechanism for Mannich reaction.	3	6	**04**